VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

VK26-IITH-SM-0180

github.com/psmithul/tensegrity-rover

Problem Context

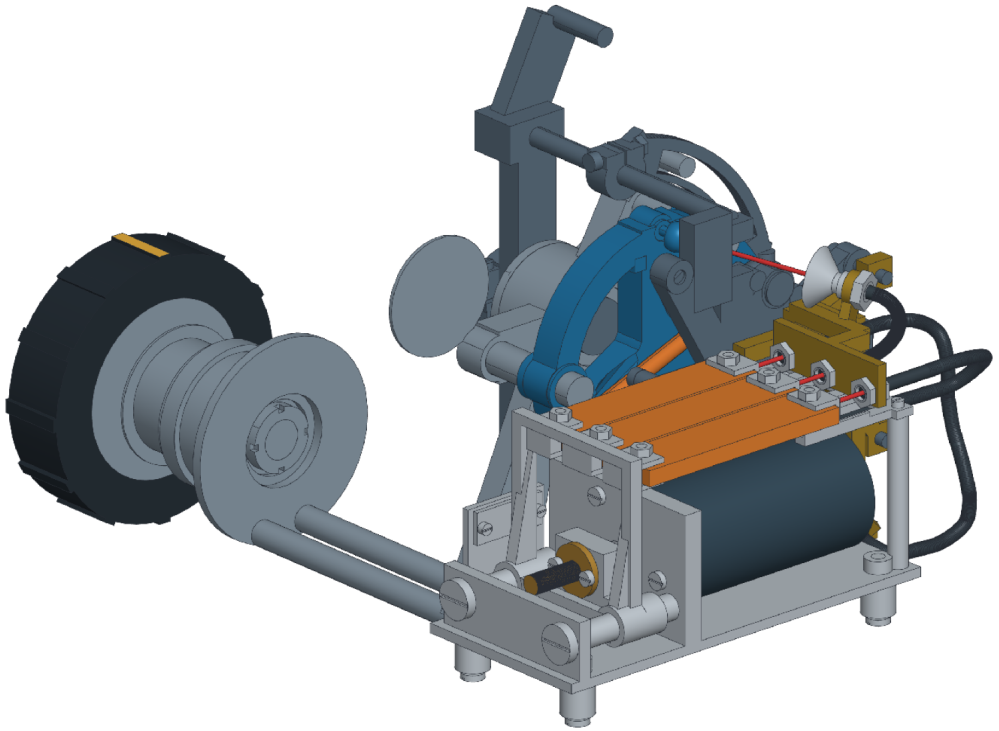
Project overview

Adaptive tensegrity suspension for rough-terrain inspection

Team ID: VK26-IITH-SM-0180
Track: Sustainable Mining
Autonomous Subterranean & Rough-Terrain Exploration
Faculty mentor: Sanjeevi Nakka

Problem

Rough ground shakes the cameras and electronics carried by an inspection rover. Slowing down can reduce excitation, but increases the time needed to cover a route.



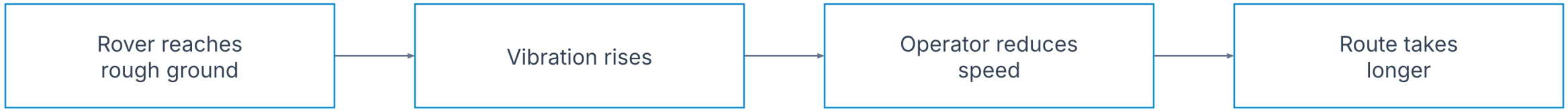
CAD of one complete suspension joint [AI Edited Image]

Target users: mine-inspection teams and rough-terrain robotics developers.

Core innovation: change joint preload using measured vibration, then select a suitable travel speed.

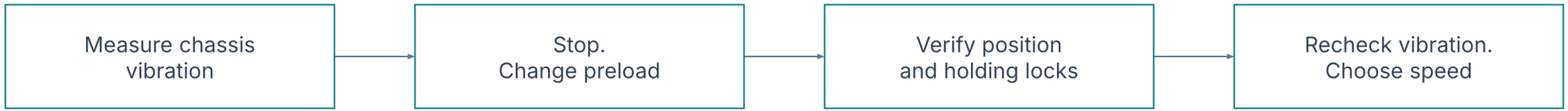
User Journey

Before: fixed suspension



Illustrative travel time: 10 m takes 100 s at 0.10 m/s, or 200 s after slowing to 0.05 m/s.

After: vibration-aware preload and speed control



Test goal: at least 20% lower chassis RMS acceleration at matched speed than fixed preload.

System Architecture & Specs

Mechanical structure
520 mm chassis, 6 suspension joints
83 mm wheels

Teensy 4.0 + Pico RP2040
Vibration, logging and drive control

3S LiPo, 11.1 V nominal
Regulation, isolation and E-stop

3 ADXL345 + MPU6050
Chassis and left/right middle carriers

6 DS3225 preload servos
6 PCA9685 command boards

4 Johnson 300 rpm drives
4 BTS7960, 2 passive wheels

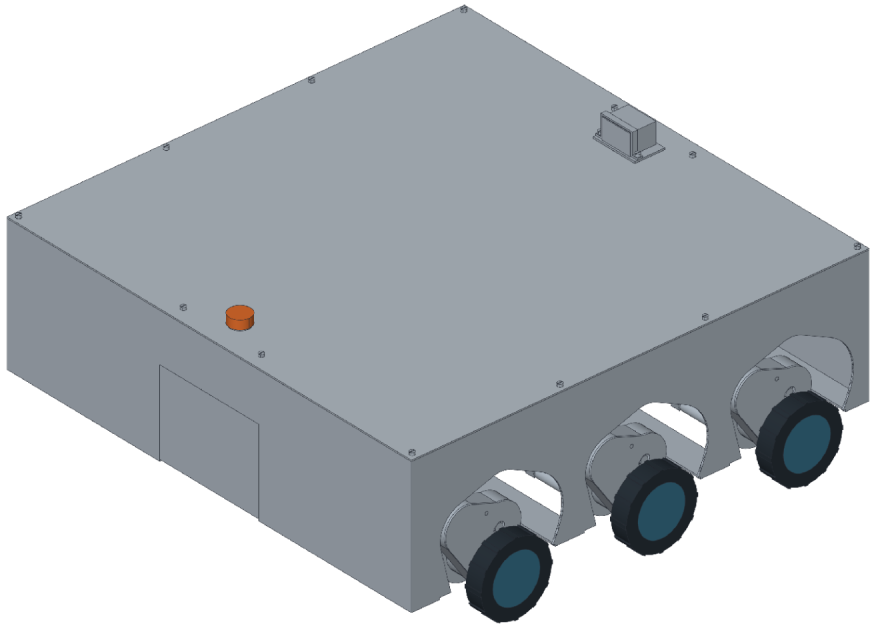
Teensy handles vibration and preload commands. Pico controls four drive motors. Stop before adjusting preload, verify the holding locks and then recheck vibration. ESP32-CAM provides the camera view.

Testing: log three ADXL345 sensors at 800 samples/s on the chassis and two middle carriers. Use the listed OE-28 as one reference-wheel encoder. Repeat fixed and adaptive tests at matched speed.

ADXL345

MPU6050

System Architecture & Specs



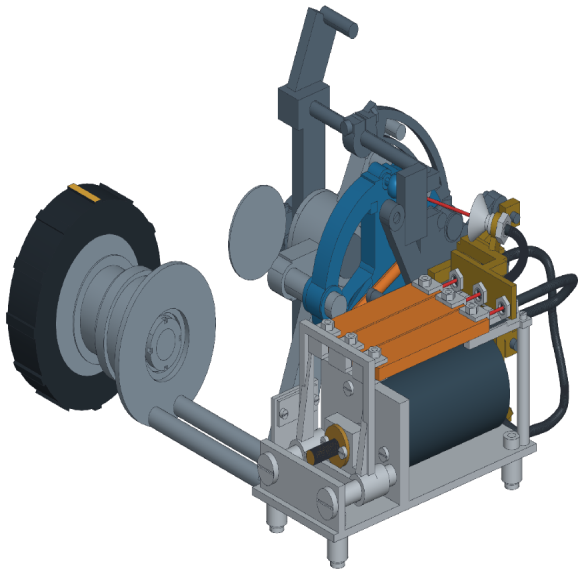
Complete rover CAD [AI Edited Image]

Preload: 6 mm travel on the M6 × 0.5 screw requires 12 turns.

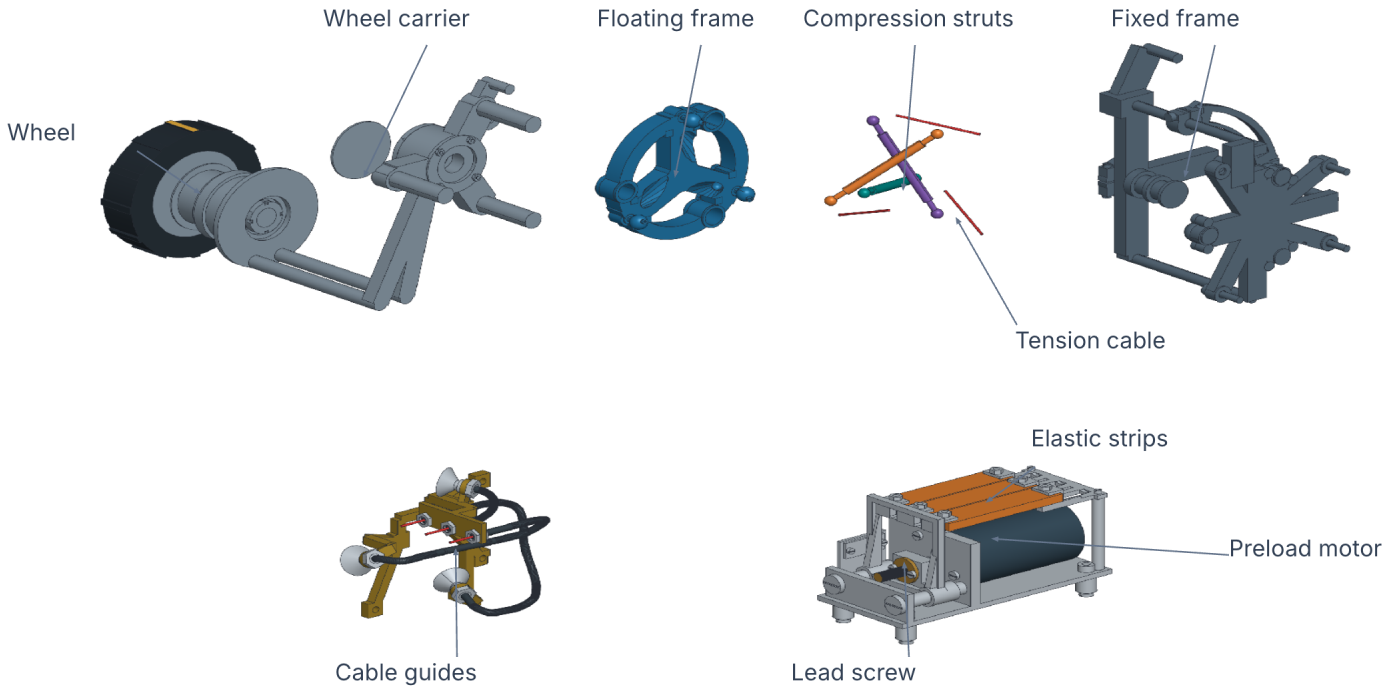
Parameter	BOM specification / design target
Total mass	30 kg design case, including battery and payload
Geometry	520 mm chassis, six 83 mm wheels
Suspension	6 passive joints, 6 DS3225 preload servos
Wheel drives	4 × Johnson 12 V, 300 rpm; 2 passive wheels
Motor control	4 × BTS7960; 6 × PCA9685
Controllers	Teensy 4.0 + Raspberry Pi Pico RP2040
Sensors / links	3 ADXL345, MPU6050; SPI, I²C and UART
Travel speed	1.304 m/s at nominal 300 rpm, before load losses
Battery / energy	2 × 11.1 V, 2.5 Ah LiPo; 27.75 Wh per pack

CAD Design & Mechanical Specs

Assembled joint



Exploded mechanical components



Load path

Rigid struts carry compression. Cable and elastic paths carry tension through supported anchors and guides.

Suspension motion

A bump rotates the carrier about the suspension pivot. Struts reorient and elastic paths extend; wheel spin drives travel.

Variable stiffness

Six servos are planned for preload adjustment. The screw-travel target remains 6 mm. Verify the output rotation, linkage and positive holding lock.

CAD Design & Mechanical Specs

Preliminary load screening

Load case	Calculation
30 kg rover weight	30 × 9.81 = 294.30 N
Six equal wheel contacts	294.30 / 6 = 49.05 N per wheel
Three equal wheel contacts	294.30 / 3 = 98.10 N per wheel
Provisional 3× static screen	3 × 98.10 = 294.30 N per wheel
65 mm perpendicular lever	294.30 × 0.065 = 19.13 Nm
Solid Ø12 mm, bending only	$32M/(\pi d^3)$ = 112.8 MPa

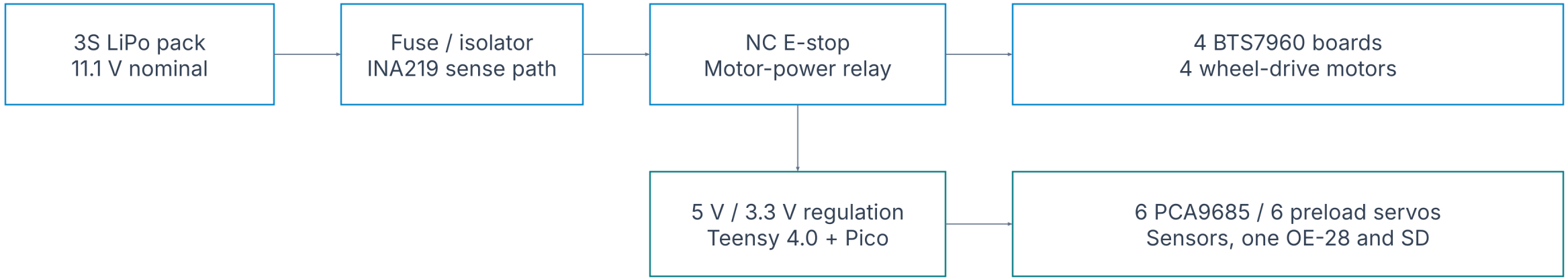
Critical locations: suspension-pivot shaft, bearing seats, ball-link ends, cable anchors and preload-nut supports.

Material and manufacturing choices

Part group	Proposed route / reason
Loaded frames and plates	Flat-cut aluminium, final grade verified before release
Shafts, hubs and bearing seats	Metal shafts and hubs. Adapt thrust seats to 51100 bearings, 10 × 24 × 9 mm.
Trays, covers and sensor mounts	PETG printing for low loads and repairability
Elastic elements / tyre trials	TPU or cut elastomer, force/creep tests required

Stock: 3 kg PETG and 1 kg TPU. Aluminium: 7.128 kg sheet/plate + 1.018 kg hub bar = 8.146 kg raw stock.

Electronics Schematics & BOM



Interface	Pin / channel assignment
Wheel drives / encoder	Pico GP2–9: four RPWM/LPWM pairs. OE-28 A/B: GP10/11.
Controller link / vibration SPI	Pico UART0 GP0/1 to Teensy Serial1 0/1. SPI 13/11/12; ADXL CS2/3/4, SD CS5.
Preload servo commands	Six PCA9685: 0x40, 0x42–0x46. CH0 on each board, 50 Hz, one servo per joint.
I²C, limits and locks	Teensy SDA18/SCL19. MCP0x20/21, MPU0x68, INA0x41. 12 NC limits + 6 lock inputs.
Analog feedback / disable	Teensy A0/A1: 8 driver diagnostics + 6 Hall inputs. Pins 6/7: driver buffer / PCA disable.

Electronics Schematics & BOM

Supplied BOM total: ₹44,069.06

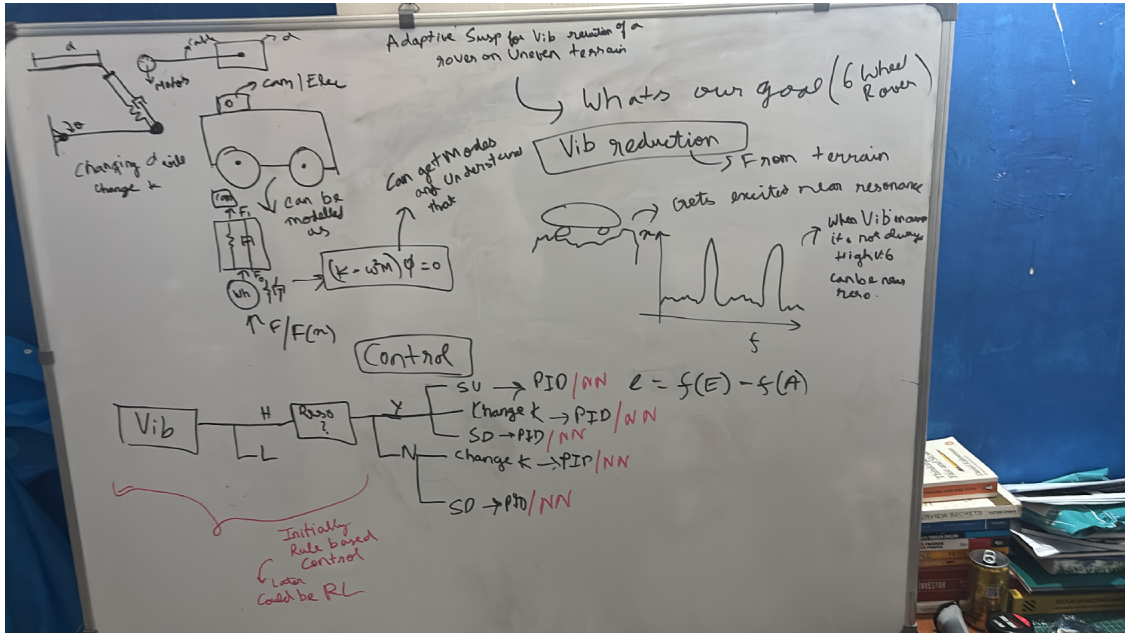
Full source BOM

Part / model	Qty	Unit ₹	Line ₹	Supplier	Delivery
Johnson 12 V, 300 rpm motor	4	583.00	2,332.00	Robu	RFQ
DS3225 preload servo	6	1,409.00	8,454.00	Robu	RFQ
PCA9685 servo driver	6	249.00	1,494.00	Robu	RFQ
BTS7960 H-bridge driver	4	329.00	1,316.00	Robu	RFQ
Raspberry Pi Pico RP2040	1	384.00	384.00	Robu	RFQ
Teensy controller	1	2,785.00	2,785.00	Robocraze	RFQ
ADXL345 accelerometer module	3	255.00	765.00	Robocraze	RFQ
MPU6050 sensor module	1	173.00	173.00	Robokits	RFQ
Encoder circuit set	1	300.00	300.00	Unquoted	RFQ
74HCT245 buffer	4	149.00	596.00	Robu	RFQ
MCP23017 I/O expander	2	332.00	664.00	FlyRobo	RFQ
CD74HC4067 analog mux	2	50.00	100.00	Robu	RFQ
Input conditioning circuit set	1	400.00	400.00	Unquoted	RFQ
Linear Hall sensor module	6	40.00	240.00	Robu	RFQ
KW11 NC limits, pack of 2	6	29.00	174.00	Robu	RFQ
SPI microSD reader	1	40.00	40.00	FlyRobo	RFQ
16 GB Class 10 microSD	1	499.00	499.00	Hubtronics	RFQ
Li-Po battery	2	1,700.00	3,400.00	Robu	RFQ
13.8 V VRLA charger	1	249.00	249.00	MakerBazar	RFQ
EF40-1A-12V power relay	1	76.70	76.70	Rajiv Elec.	RFQ
Relay connection set	1	70.00	70.00	Rajiv/local	RFQ
NC emergency-stop switch	1	109.00	109.00	Robu	RFQ

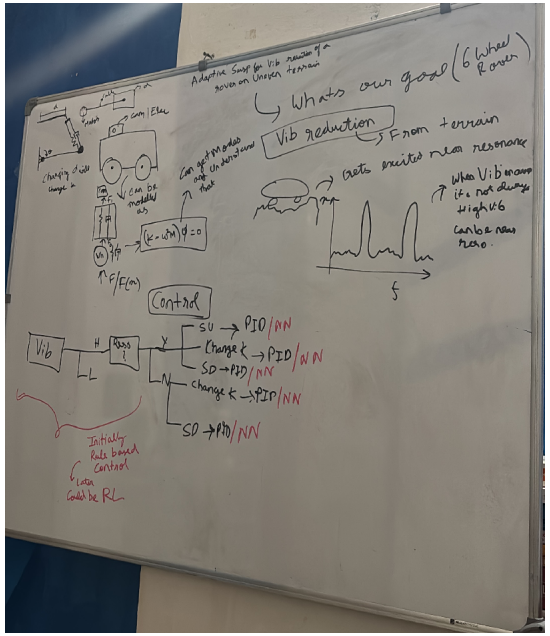
Part / model	Qty	Unit ₹	Line ₹	Supplier	Delivery
K800 ring magnet	1	29.00	29.00	Robu	RFQ
5 V DC/DC regulator	1	93.00	93.00	FlyRobo	RFQ
3.3 V regulator set	1	50.00	50.00	Unquoted	RFQ
ESP32-CAM camera	1	750.00	750.00	Robocraze	RFQ
Power harness set	1	900.00	900.00	Robu/local	RFQ
Signal harness set	1	700.00	700.00	ProBots	RFQ
NC lock switches, pack of 2	3	29.00	87.00	Robu	RFQ
Battery energy measurement kit	1	139.00	139.00	Robu	RFQ
OE-28 rotary encoder	1	500.00	500.00	Robu	RFQ
Aluminium sheet and plate	1	3,304.00	3,304.00	Rajesh Al.	RFQ
Aluminium Ø40×300 mm bar	1	708.00	708.00	Rajesh Al.	RFQ
Steel and brass stock	1	1,534.00	1,534.00	India stock	RFQ
40×20×2 mm lock tube	2	206.50	413.00	Manish St.	RFQ
Polymer stock	1	1,062.00	1,062.00	Robu/local	RFQ
PETG 1 kg spool	3	798.86	2,396.58	DreamPoly.	RFQ
TPU 95A 1 kg spool	1	1,414.82	1,414.82	3D Master	RFQ
Tension cable and fittings	1	944.00	944.00	EBMP India	RFQ
Radial bearing 12×28×8 mm	24	21.00	504.00	Omrook	RFQ
51100 thrust bearing	6	102.66	615.96	Omrook	RFQ
M2-M6 fasteners and inserts	1	2,832.00	2,832.00	Local stock	RFQ
Assembly consumables	1	472.00	472.00	Joy Guru	RFQ

Maker Journey

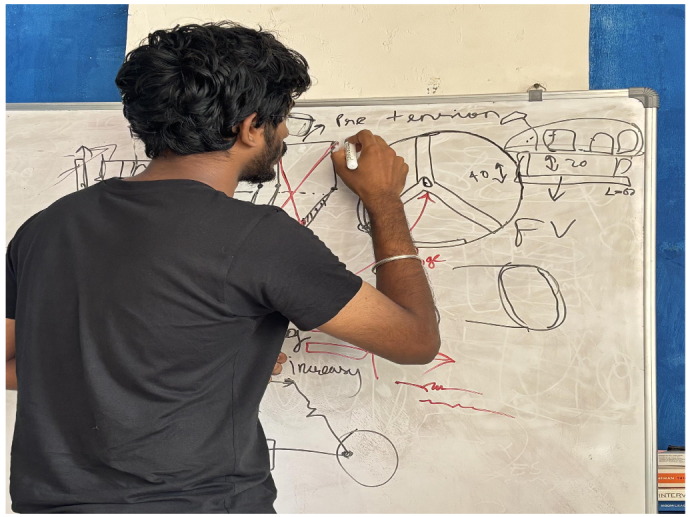
From the vibration problem to a suspension concept



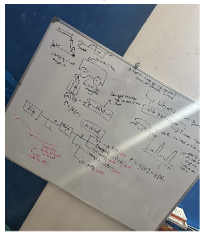
Identify vibration and resonance



Compare speed and stiffness



Identify vibration and resonance

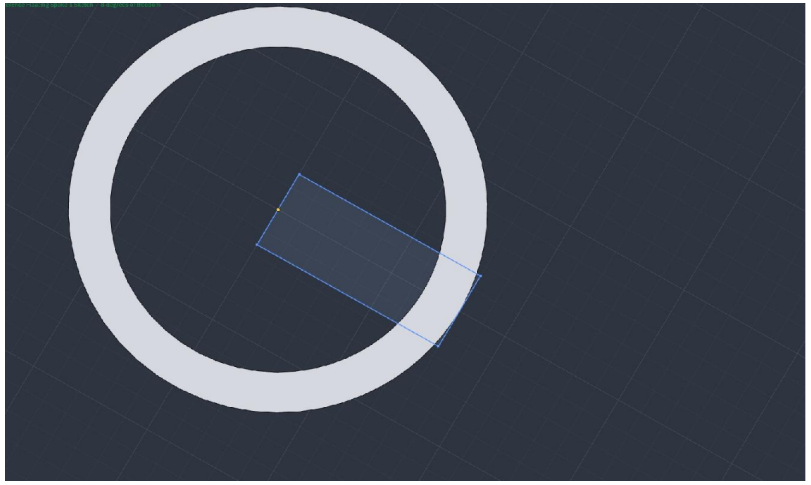


Sketch preload and wheel loads

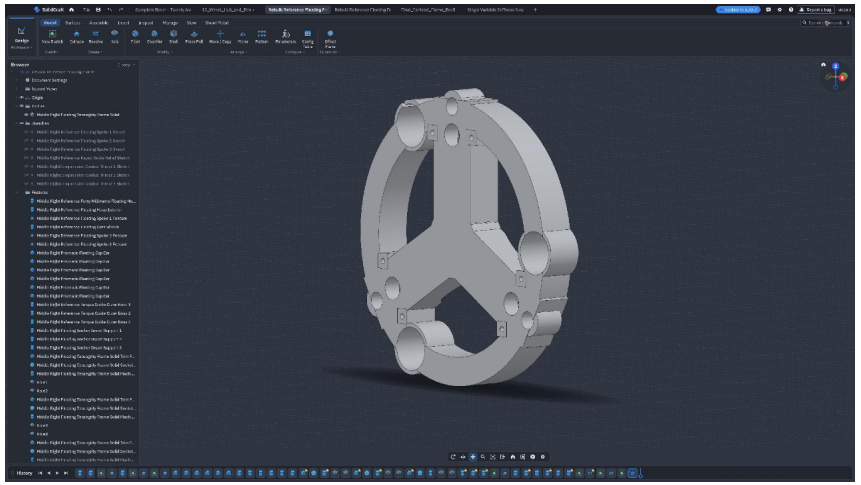
We first asked whether to slow the rover down or change the suspension response. The sketches led us to adjustable preload and a simple rule-based controller. The next step is to measure the joint response before tuning the controller.

Maker Journey

Floating frame: sketch to finished part



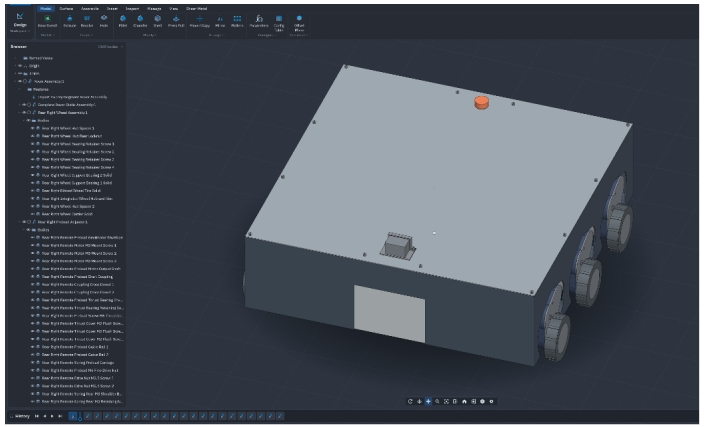
Original spoke sketch



Finished frame and feature tree

Frame modelling

Original spoke sketch



Final model Feature Tree

Complete rover CAD review

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[Raw STEP models]

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Safety & Hazard Mitigation

Failure mode	Hazard	Physical / software protection
Cable or elastic element breaks	Loss of elastic support	Retained ends, hard stops and residual-support load test
Preload servo loss, stall or jam	Preload drift, overtravel or heating	Hard stops, 12 NC end limits and 6 lock inputs. Qualify servo supply and positive holding locks.
IMU / logger fault or lost radio	Unsafe control command	Watchdog stops PWM and enters a stopped state
Wheel stall / H-bridge heat	Overcurrent and overheating	Qualify four drive branches for fusing, current cutoff and thermal protection. Test startup and stall loads.
Battery or wiring fault	Fire or electric damage	Use a compatible 3S LiPo balance charger and discharge protection. The listed VRLA charger is incompatible.

Physical fail-safes

We will fit an NC emergency stop to cut wheel and servo power independently of firmware, and test manual reset, holding locks and guarded access.

Design guidelines considered

ISO 12100, ISO 13850 and IEC 60529. Bench-check safe reset, power loss and all six holding locks before loaded travel.

Roadmap

Build and qualify one joint before repeating it six times

Month	Build and integration work	Exit gate
1	Confirm servo rotation, 6 mm travel, 51100 seats and power/charger interfaces. Make one joint.	Demonstrate travel, locking, force response and temperature under staged loads
2	Build six joints, four driven and two passive wheels. Fit 4 BTS7960, 6 PCA9685 and the controllers.	Verify E-stop, end limits, locks, converter margin and staged static loads
3	Calibrate 3 ADXL345, MPU6050 and the listed encoder. Qualify Hall feedback and implement control.	Reliable logs, repeatable preload settings and safe fault handling
4	Repeat fixed and adaptive tests on the same rough track, speed and payload.	Compare vibration, total route time and energy. Retain the raw logs

Stage 3 target: a working rover ready for a controlled rough-terrain demonstration, supported by repeatable test data.

Technical Challenges & Risks

Constraint	Evidence / current limitation
30 kg in a compact chassis	294.3 N provisional wheel screen and 19.13 Nm arm moment.
Servo and bearing interfaces	6 mm screw travel requires 12 turns. Confirm servo output rotation. The selected 51100 thrust bearings need revised seats.
Stiffness, friction and damping	Preload response depends on elastic force and guide friction. No measured stiffness or damping curve yet.
Drive and feedback coverage	300 rpm gives 1.304 m/s nominal at 83 mm. Creep needs 11.5–23.0 rpm. One encoder and one magnet are listed.
Battery and servo power	Six servos can demand 11.4 A at 5 V stall against a listed 5 A converter. The VRLA charger is incompatible with 3S LiPo.

Potential Solutions & Risk Mitigation

Risk	Primary action	Fallback
Concentrated load / print creep	Metal shafts, loaded frames and bearing seats. Qualify the new 51100 seats and one loaded joint.	Increase section or replace critical printed parts with flat metal
Cable friction or wear	Measure tension loss and sweep travel at each preload. Use supported direction changes.	Increase guide radius or use a rated pulley
Servo travel or holding fails	Verify 12 screw turns for 6 mm travel. Unload one joint, adjust and confirm the positive lock.	Revise the servo output or transmission after measuring torque and stroke
Low-speed drive / limited feedback	Test four motors at 11.5–23.0 wheel rpm. Use one reference encoder and repeated timed track runs.	Review reduction and encoder coverage before speed-control release
Power and charger integration	Match a 3S LiPo charger and qualify servo current, regulator cooling, shunt and emergency power removal.	Keep fixed preload and hold loaded tests until the power design passes

References & Sources

Research and background sources

O. M. Anubi, D. R. Patel and C. D. Crane III. (2013). A new variable stiffness suspension system: passive case.
Mechanical Sciences, 4, 139–151.

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D. Zappetti, Y. Sun, M. Gevers, S. Mintchev and D. Floreano. (2022).
Dual Stiffness Tensegrity Platform for Resilient Robotics. Advanced Intelligent Systems, 4(7), 2200025.

https://doi.org/10.1002/aisy.202200025

S. Lessard, J. Bruce, E. Jung, M. Teodorescu, V. SunSpiral and A. Agogino. (2015).
A light-weight, multi-axis compliant tensegrity joint. arXiv:1510.07595v1.

https://arxiv.org/html/1510.07595v1

ScienceDirect Topics. Tensegrity - an overview. Engineering topic reference.

https://www.sciencedirect.com/topics/engineering/tensegrity

Project sources: team proposal, CY06 drawings, CAD, screenshots and development photos. Sensor photos: RoboticsDNA. Technical sources: ADXL345, MPU6050, RP2040, Teensy, PCA9685, MCP23017, INA219 and DS3225 documentation. Costs: supplied 43-line BOM.

github.com/psmithul/tensegrity-rover

Sensor module sources: RoboticsDNA

AI assistance was used to prepare this presentation and generate some images without backgrounds.